

Siyuan Mei

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Pattern Recognition Lab (LME), FAU Erlangen–Nürnberg, Germany

RESEARCH INTERESTS

- My research focuses on **medical AI and computer vision**, particularly medical image generation, reconstruction, segmentation, and registration, with methodological interests in diffusion and flow models, representation learning, foundation models, and agentic AI systems. Going forward, I aim to translate these methods into robust and practical AI systems.

EDUCATION

- **FAU Erlangen–Nürnberg** Sep. 2023–May 2027 (expected)
Fully funded Ph.D. in Computer Science; Supervisor: Prof. Dr.-Ing. habil. Andreas Maier Erlangen, Germany
- **Peking University Health Science Center** May 2026–Oct. 2026
Visiting Scholar; Supervisor: Prof. Dr.-Ing. Yixing Huang Beijing, China
- **FAU Erlangen–Nürnberg** Sep. 2020–June 2023
M.Sc. in Medical Engineering; Thesis grade: 1.0 Erlangen, Germany
- **Zhejiang University of Science and Technology** Sep. 2016–June 2020
B.Eng. in Automation; Sino-German 2+3 Program; Honored Graduate Scholarship Hangzhou, China

PROFESSIONAL EXPERIENCE

- **Friedrich-Alexander-Universität Erlangen–Nürnberg (FAU)** 2023–2027
Ph.D. Student, Pattern Recognition Lab; Supervisor: Prof. Andreas Maier Erlangen, Germany
 - **Fully funded** through the **European Research Council (ERC) project** *Advancing osteoporosis medicine by observing bone microstructure and remodelling using a 4D nanoscope*.
 - Developed **BigReg**, an efficient registration pipeline for high-resolution X-ray microscopy and light-sheet fluorescence microscopy.
 - Conducted research in medical AI and computer vision, adapting foundation models and developing generative methods based on GANs, diffusion models, and flow matching.
 - Long-term experience training AI models on large-scale H100/A100 clusters, familiar with model parallelism and distributed frameworks.
 - Led the experimental and writing work for the team achieving **1st place in the MICCAI SynthRAD2025 Challenge**.
 - Taught *Flat-panel CT Reconstruction* and contributed lecture slides for *Vibe Coding: Software Engineering with LLMs*, a Springer book project.
 - **Research outcomes:** First-author publications in **Pattern Recognition**, **MICCAI 2026 (Oral)**, **ISBI 2023**, **CT Meeting 2024**, and **JMI (Featured)**, together with seven co-authored **MICCAI** papers and journal publications in **TMI**, **PMB**, **BMPC**.
- **Peking University, Artificial Intelligence for Medical Imaging (AIMI) Lab** May–Oct. 2026
Visiting Scholar; Supervisor: Prof. Yixing Huang Beijing, China
 - Participated in the MICCAI BIC-MAC Challenge and trained a multimodal-to-CT generative model for PET attenuation correction; achieved **first place on the validation-set CT metric**.
 - Expanded practical knowledge of clinical CBCT/CT applications and imaging workflows through research visits and academic exchange, including the ICT Center in Chongqing and MISC 2026 at Tsinghua University.

SELECTED PUBLICATIONS

[J=Journal](#), [C=Conference](#), [M=Manuscript](#)

- [C.1] **Siyuan Mei**, Yanteng Zhang, Yan Xia, Qizhen Lan, Yipeng Sun, Siming Bayer, Zirong Li, Chengze Ye, Daiqi Liu, Xiaoqian Jiang, Fuxin Fan, Yixing Huang, and Andreas Maier. *Time Matters: Rethinking Diffusion and Flow Models in One-Step Medical Image Translation*. **MICCAI, 2026 (Early Accept; Oral Presentation)**.

- [C.2] **Siyuan Mei**, Yan Xia, Yipeng Sun, Siming Bayer, Zirong Li, Chengze Ye, Daiqi Liu, Fuxin Fan, Yixing Huang, and Andreas Maier. *WING: A Window-Prior-Based Generative Network with Gated Inception for Cross-Modality CT Synthesis*. **MICCAI, 2026** (Accepted workshop paper).
- [C.3] Daiqi Liu, Lukas Mulzer, Md Hasan, Nyvenn de Castro, Fangxu Xing, Xingjian Kang, Chengze Ye, **Siyuan Mei**, Yipeng Sun, Tomás Arias-Vergara, Jana Hutter, Jonghye Woo, Andreas Maier, and Paula Andrea Pérez-Toro. *Speech-Guided Multimodal Learning for Vocal Tract Segmentation in Real-Time MRI*. **MICCAI, 2026** (**Early Accept; Oral Presentation**).
- [C.4] Mengchen Fan, Baocheng Geng, Xi Xiao, Tianyang Wang, **Siyuan Mei**, Pulin Che, Xiaoqian Jiang, and Qizhen Lan. *Detail Consistent Stage-Wise Distillation for Efficient 3D MRI Segmentation*. **MICCAI, 2026** (**Early Accept; Spotlight Presentation**).
- [J.1] **Siyuan Mei**, Mareike Thies, Yan Xia, Yipeng Sun, Fei Wu, Fuxin Fan, Mingxuan Gu, Chengze Ye, Yixing Huang, Vincent Christlein, and Andreas Maier. *Vision Transformer Hook for Dense Predictions*. **Pattern Recognition, 179** (2026), 113818.
- [J.2] **Siyuan Mei**, Fuxin Fan, Mareike Thies, Mingxuan Gu, Fabian Wagner, Oliver Aust, Ina Erceg, Zeynab Mirzaei, Georgiana Neag, Yipeng Sun, Yixing Huang, and Andreas Maier. *BigReg: An Efficient Registration Pipeline for High-Resolution X-Ray and Light-Sheet Fluorescence Microscopy*. **Journal of Medical Imaging, 12(5), 054004** (2025) (**Featured Paper**).
- [C.5] **Siyuan Mei**, Fuxin Fan, and Andreas Maier. *Metal Inpainting in CBCT Projections Using Score-based Generative Model*. **IEEE ISBI, 2023**.
- [J.3] Mareike Thies, Fabian Wagner, Noah Maul, Haijun Yu, Manuela Goldmann, Linda-Sophie Schneider, Mingxuan Gu, **Siyuan Mei**, Lukas Folle, Alexander Preuhs, Michael Manhart, and Andreas Maier. *A Gradient-Based Approach to Fast and Accurate Head Motion Compensation in Cone-Beam CT*. **IEEE Transactions on Medical Imaging, 44(2), 1098–1109** (2025).
- [J.4] Chengze Ye, Linda-Sophie Schneider, Yipeng Sun, Mareike Thies, **Siyuan Mei**, Paula Andrea Pérez-Toro, Siming Bayer, and Andreas Maier. *Robustness and Stability Analysis of Differentiable Shift-Variant FBP for Cone-Beam CT under Challenging Acquisition Settings*. **Machine Learning for Biomedical Imaging, 2026, 284–296**.
- [J.5] Yipeng Sun, Linda-Sophie Schneider, **Siyuan Mei**, Jinhua Wang, Ge Hu, Mingxuan Gu, Chengze Ye, Fabian Wagner, Lan Song, Siming Bayer, and Andreas Maier. *Filter2Noise: A Framework for Interpretable and Zero-Shot Low-Dose CT Image Denoising*. **Journal of Medical Imaging, 13(2), 024004** (2026).
- [J.6] Chengze Ye, Linda-Sophie Schneider, Yipeng Sun, Mareike Thies, **Siyuan Mei**, and Andreas Maier. *DRACO: Differentiable Reconstruction for Arbitrary CBCT Orbits*. **Physics in Medicine and Biology, 70, 075005** (2025).
- [C.6] Mareike Thies, Noah Maul, **Siyuan Mei**, Laura Pfaff, Nastassia Vysotskaya, Mingxuan Gu, Jonas Utz, Dennis Possart, Lukas Folle, Fabian Wagner, and Andreas Maier. *Differentiable Score-Based Likelihoods: Learning CT Motion Compensation From Clean Images*. **MICCAI, 2024**.
- [C.7] Mingxuan Gu, Mareike Thies, **Siyuan Mei**, Fabian Wagner, Mingcheng Fan, Yipeng Sun, Zhaoya Pan, Sulaiman Vesal, Ronak Kost, Dennis Possart, Jonas Utz, and Andreas Maier. *Unsupervised Domain Adaptation Using Soft-Labeled Contrastive Learning with Reversed Monte Carlo Method for Cardiac Image Segmentation*. **MICCAI, 2024**.
- [C.8] Laura Pfaff, Fabian Wagner, Nastassia Vysotskaya, Mareike Thies, Noah Maul, **Siyuan Mei**, Tobias Wuerfl, and Andreas Maier. *No-New-Denoiser: A Critical Analysis of Diffusion Models for Medical Image Denoising*. **MICCAI, 2024**.
- [M.1] Zirong Li, **Siyuan Mei**, Weiwen Wu, Andreas Maier, Lina Götz, and Yan Xia. *Generative Drifting for Conditional Medical Image Generation*. **arXiv preprint, 2026**.
- [M.2] **Siyuan Mei**, Yan Xia, and Fuxin Fan. *GANeXt: A Fully ConvNeXt-Enhanced Generative Adversarial Network for MRI- and CBCT-to-CT Synthesis*. **arXiv preprint (Challenge Report), 2025**.

HONORS AND AWARDS

• 1st place, Task 2 and 4th place, Task 1, MICCAI SynthRAD2025 Challenge	2025
• Excellent International Graduate Student Scholarship, FAU Erlangen–Nürnberg	2023
• Excellent MRI Course Achievement Scholarship, FAU Erlangen–Nürnberg	2022
• Honored Undergraduate Scholarship, Zhejiang University of Science and Technology	2020

TEACHING

- **Flat-panel CT Reconstruction**

Summer semesters 2024–2026

FAU Erlangen–Nürnberg

- **Vibe Coding: Software Engineering with LLMs**

Summer semester 2026

FAU Erlangen–Nürnberg

ACADEMIC SERVICES

- **Journal Reviewer**

- Pattern Recognition
- Journal of Machine Learning for Biomedical Imaging (MELBA)

- **Technical Editor**

- Journal of Medical Imaging

- **Conference Reviewer**

- International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)

ADDITIONAL INFORMATION

- **Languages:** Chinese (native), English (professional), German (C1 certificate).
- **Interests:** Fitness and tennis (level 2.5).
- **MBTI:** ENTJ-A.
- **Professional interests:** Open to research collaborations and industry opportunities in medical imaging, computer vision, multimodal models, and agentic AI systems.
- **Open-source teaching resources:** [Vibe Coding lecture slides](#).